CHAPTER 1

SOME BASIC CONCEPTS OF CHEMISTRY

CHAPTER ESSENCE AT A GLANCE:
Governs quantitative chemical relationships, fundamental mass conservation laws, the mole concept, empirical formula determination, and solution concentration scales.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Mole Concept & Avogadro Number:** One mole corresponds to 6.022×10^{23} elementary entities with mass matching formula weight in grams.

• **Law of Definite Proportions:** A pure chemical compound always contains elements combined in a fixed ratio by mass regardless of source.

• **Empirical vs Molecular Formula:** Empirical formula gives simplest integer ratio; molecular formula gives actual atom counts via integer multiple n.

• **Limiting Reagent Stoichiometry:** The reactant completely consumed first limits product formation and dictates theoretical yield.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$M = \frac{w_B \times 1000}{M_B \times V_{\text{mL}}}$$

Molarity (Moles of solute per litre of solution)

$$m = \frac{w_B \times 1000}{M_B \times W_A(\text{g})}$$

Molality (Moles of solute per kilogram of solvent)

$$x_B = \frac{n_B}{n_A + n_B} \quad (x_A + x_B = 1)$$

Mole Fraction of Solute and Solvent

$$\text{Molecular Formula} = (\text{Empirical Formula})_n, \quad n = \frac{M_{\text{molecular}}}{M_{\text{empirical}}}$$

Molecular Formula Integral Multiplier

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Molarity depends on temperature because volume expands; molality is temperature independent because mass is invariant.

• **Alert:** Always balance the chemical equation and identify the limiting reagent before calculating product mass.

• **Alert:** In mole fraction calculations, verify that the sum of mole fractions of all components identically equals 1.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define limiting reagent in a chemical reaction. Explain with a suitable example. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** The reactant which gets completely consumed first in a reaction and limits the amount of product formed is called the limiting reagent. **[1 Mark]**

• **Example & Valuation Step:** In the reaction 2H2 + O2 -> 2H2O, 2 moles of H2 require 1 mole of O2. If 2 moles of H2 and 2 moles of O2 are reacted, H2 is consumed first and acts as the limiting reagent, leaving 1 mole of O2 in excess. **[1 Mark]**

• **Key Exam Result:** Limiting reagent determines the theoretical yield of products in chemical stoichiometry.

2. Differentiate between Molarity and Molality of a solution. Why is molality preferred over molarity in temperature-dependent studies? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Comparison of Definitions:** Molarity (M) is the number of moles of solute dissolved in one litre of solution, whereas Molality (m) is the number of moles of solute present in one kilogram of solvent. **[1 Mark]**

• **Temperature Dependence:** Molarity involves solution volume which expands or contracts with temperature changes, altering molar concentration. Molality involves solvent mass which is invariant with temperature, making molality preferable. **[1 Mark]**

• **Key Exam Result:** Molality is temperature-independent; Molarity varies inversely with thermal expansion.

3. An organic compound contains 4.07% hydrogen, 24.27% carbon, and 71.65% chlorine. Its molar mass is 98.96 g/mol. Determine its empirical formula and molecular formula. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Mole Ratio Calculation):** Moles of C = 24.27 divided by 12.01 = 2.021. Moles of H = 4.07 divided by 1.008 = 4.038. Moles of Cl = 71.65 divided by 35.45 = 2.021. **[1 Mark]**

• **Step 2 (Simplest Whole Number Ratio):** Dividing by 2.021 gives C : H : Cl = 1 : 2 : 1. Empirical formula = CH2Cl. Empirical formula mass = 12 + 2(1) + 35.5 = 49.5 g/mol. **[1 Mark]**

• **Step 3 (Molecular Formula):** Integral factor n = Molar mass divided by Empirical mass = 98.96 divided by 49.5 = 2. Molecular formula = (CH2Cl)2 = C2H4Cl2. **[1 Mark]**

• **Key Exam Result:** Empirical formula = CH2Cl; Molecular formula = C2H4Cl2 (Dichloroethane).

$$\text{Empirical: } \mathrm{CH_2Cl}, \quad n = \frac{98.96}{49.48} \approx 2 \rightarrow \mathrm{C_2H_4Cl_2}$$

Empirical and Molecular Formula Evaluation

4. 50.0 kg of N2 (g) and 10.0 kg of H2 (g) are mixed to produce NH3 (g). Identify the limiting reagent and calculate the mass of NH3 formed. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Moles of Reactants):** Moles of N2 = 50000 g divided by 28 g/mol = 1785.7 mol. Moles of H2 = 10000 g divided by 2.016 g/mol = 4960.3 mol. **[1 Mark]**

• **Step 2 (Limiting Reagent Identification):** Balanced reaction is N2 + 3H2 -> 2NH3. 1785.7 moles of N2 require 3 × 1785.7 = 5357.1 moles of H2. Since only 4960.3 moles of H2 are available, H2 is the limiting reagent. **[1 Mark]**

• **Step 3 (Product Yield & Boxed Result):** 3 moles of H2 yield 2 moles of NH3. Moles of NH3 = (2 divided by 3) × 4960.3 = 3306.9 mol. Mass of NH3 = 3306.9 mol × 17 g/mol = 56217 g = 56.1 kg. **[1 Mark]**

• **Key Exam Result:** Limiting reagent is H2; Mass of NH3 produced = 56.1 kg.

$$\text{Moles } \mathrm{NH_3} = \frac{2}{3} \times n_{\mathrm{H_2}} = 3306.9 \text{ mol} \rightarrow \text{Mass} = 56.1 \text{ kg}$$

Stoichiometric Yield Calculation

5. Calculate the molarity of commercial nitric acid solution which has 69% by mass HNO3 and density of 1.41 g/mL. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Mass of Solute in 1 Litre):** 69% by mass means 100 g solution contains 69 g HNO3. Volume of 100 g solution = mass divided by density = 100 divided by 1.41 = 70.92 mL. **[1 Mark]**

• **Step 2 (Molar Mass and Moles):** Molar mass of HNO3 = 1 + 14 + 3(16) = 63 g/mol. Moles of HNO3 in 100 g solution = 69 divided by 63 = 1.095 mol. **[1 Mark]**

• **Step 3 (Molarity Calculation):** Molarity = (moles of solute × 1000) divided by volume in mL = (1.095 × 1000) divided by 70.92 = 15.44 mol/L. **[1 Mark]**

• **Key Exam Result:** Molarity of 69% nitric acid solution = 15.44 mol/L.

$$M = \frac{\% \times d \times 10}{M_B} = \frac{69 \times 1.41 \times 10}{63} = 15.44 \text{ M}$$

Concentrated Acid Molarity Formula

6. State and explain Gay Lussac's Law of Gaseous Volumes and Avogadro's Law. How did Avogadro reconcile Dalton's atomic hypothesis? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Gay Lussac's Law:** When gases react together, they do so in volumes which bear a simple whole number ratio to one another and to the volumes of gaseous products, measured at identical temperature and pressure. **[1 Mark]**

• **Avogadro's Law:** Equal volumes of all gases under identical conditions of temperature and pressure contain an equal number of molecules (V is proportional to n). **[1 Mark]**

• **Reconciliation with Dalton:** Avogadro distinguished between atoms and molecules, proving that elementary gases like hydrogen and oxygen are diatomic, avoiding fractional atoms in Dalton's model. **[1 Mark]**

• **Key Exam Result:** Avogadro's hypothesis successfully integrated Gay Lussac's gas volume observations with atomic theory.